

Question	Answer	Marks
6(a)	plate X marked as negative and plate Y marked as positive	B1
6(b)(i)	$E = V / x$	C1
	$= (58 \times 10^3) / 0.041$ $= 1.4 \times 10^6 \text{ N C}^{-1}$	A1
6(b)(ii)	$ma = eE$	C1
	$a = (1.60 \times 10^{-19} \times 1.41 \times 10^6) / (9.11 \times 10^{-31})$ $= 2.5 \times 10^{17} \text{ m s}^{-2}$	A1
6(c)(i)	$eV = hc / \lambda$ or $eV = hf$ and $f = c / \lambda$	C1
	$(1.60 \times 10^{-19} \times 58 \times 10^3) = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$	M1
	clear conversion from m to pm leading to $\lambda = 21 \text{ pm}$	A1
6(c)(ii)	X-rays	B1
6(c)(iii)	<i>Any two points from:</i> - waves are passed into structure and transmitted waves detected - different parts of the structure absorb different fractions of energy - difference in detected / transmitted <u>intensities</u> used (to form image)	B2

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